ONTOLOGY MAPPING

Using the OxO mapping service

Finding, reading, and trusting mappings between ontology terms.

A guide to the OxO2 web interface · last updated 2026-11-13. If a control described here is missing from your instance, the instance is older than this document.

> OxO answers one question: *this term over here — what is it called over there?* You give it a term, it gives you the mappings other people have already published, tells you who published each one and how confident they were, and gets out of the way.

OxO API Documentation

Everything this interface does, it does over a read-only REST API — and that API describes itself. Relative to whatever URL your OxO instance is served from:

- `/swagger-ui.html` — Swagger UI: every endpoint with its parameters and responses, and a form to call it against that instance.
- `/v3/api-docs` — the same thing as a raw OpenAPI 3 specification, for generating a client.

The **Documentation** tab of the instance you are using links to both directly, so follow them from there rather than typing a URL — every deployment sits at its own address, and the tab always points at its own. There is no write API: mappings are loaded by an offline pipeline, so every endpoint only reads.

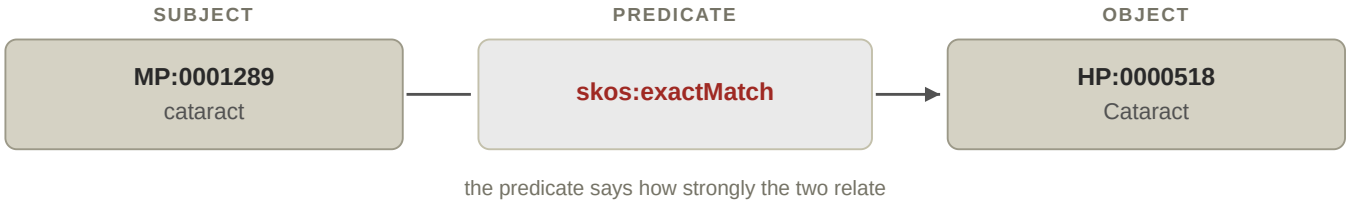
The rest of this document is about the web interface.

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1 What OxO does

A **mapping** is a statement that one term means (roughly) the same thing as another. OxO stores mappings in the SSSOM format, where every mapping is a sentence with three parts:



the predicate says how strongly the two relate

Every row in an OxO result is one of these sentences.

OxO holds mappings from two kinds of source, and the distinction matters when you decide what to trust:

- **Ontologies.** Cross-references an ontology asserts about its own terms — the matches the ontology maintainers put there themselves.
- **Curated sets.** Mappings published as standalone SSSOM files, such as EVORA or Mapping Commons, usually the output of a deliberate curation effort.

On top of both, OxO derives new mappings by chaining existing ones: if `A exactMatch B` and `B exactMatch C`, then `A exactMatch C`. These are **inferred** mappings, marked SSSOM inference throughout the interface, as opposed to Asserted ones that somebody actually wrote down. An inferred mapping is a real answer, but it is a derived one, and OxO can always show you the chain it came from (§ 7).

2 The screens

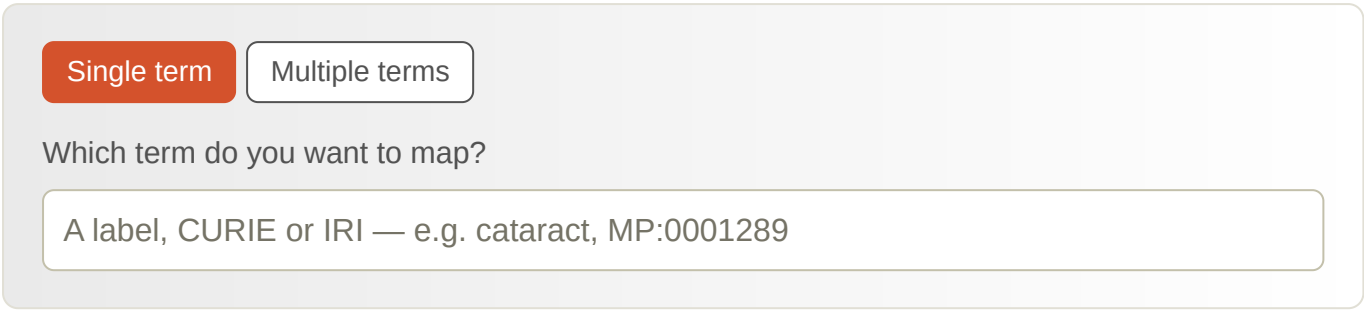
Screen	Address	What it is for
Home	`/`, `/home`	The search form, plus cards linking to About, related tools, and the issue tracker.
Results	`/search/<terms>`	The search form again (so you can refine without going back), then the results table.
Whole-ontology results	`/search/_map?from=…&to=…`	What a search with ontologies but no terms produces. `_map` is a placeholder meaning “no terms”.
Mapping details	`/mapping/<id>`	Everything SSSOM records about one mapping, and — if it is inferred — the graph explaining it.
Mapping set details	`/mapping-set?id=<set IRI>`	One set’s metadata and all of its mappings. Reached by clicking any mapping set name.
Inferred sets	`/inferences`, `/inferences/<source>`	The same page as above, resolving OxO’s own inferred sets by their published IRIs.
About	`/about`	What OxO is and who maintains it.
Documentation	`/docs`	This guide, shown in the page — with links to open it in its own tab or download it as a PDF.

The search form on the Home page and the one above the results are the same component. Refining a search never means navigating backwards.

3 Searching

There is one search form and it asks one question, with everything else folded behind **More options**. The **Search** and **Clear** buttons sit *above* that disclosure, so they do not jump down the page when you open it.

3.1 One term, or many



Single-term mode. Press `Enter` to search.

Single term gives you one input box with a suggestion list (§ 3.3); `Enter` searches. **Multiple terms** swaps it for a text area — one term per line — plus a file drop and a running *"n terms to map"* count. In multiple-term mode `Enter` starts a new line, so use the **Search** button.

Both modes have an **Example…** link that fills the box with something that works. Use it once and you will never need this section again.

3.2 What you can type

OxO recognises three kinds of term, and you may mix them freely in one search:

Kind	Example	How it is matched
Label	`cataract`	Against term labels. Exactly, ignoring case, unless you change it (§ 3.6).
CURIE	`MP:0001289`	Exact identifier lookup. Always exact, whatever the label-matching setting says.
IRI	`http://purl.obolibrary.org/obo/UBERON_0002107`	Exact IRI lookup. Also always exact.

Terms are separated by newlines or commas, and surrounding whitespace is trimmed.

> **WHY LABEL SEARCH IS EXACT BY DEFAULT**
>
> A partial search for `cataract` also returns *"congenital cataract"*, *"cataract surgery"*, and several hundred others — which buries the term you actually meant. Exact-but-case-insensitive is almost always what you want. Switch to **Partial match** under *More options* when you are exploring rather than looking something up.

> **YOUR TERM IS MATCHED ON THE SUBJECT SIDE**
>
> A search finds mappings where your term is the *subject*, not the object. A mapping is a symmetric fact but a table is not: showing both sides would list every term twice, once in each column. If you want the reverse direction, search the other term.

3.3 The suggestion list

Typing two or more characters into the single-term box opens a list of up to ten matching terms. Each row shows the term's **label**, then its **CURIE**, then its **IRI**, with the number of mappings it will return on the right. Picking one **runs the search immediately** — it does not just fill the box.

The count is the point. It is not the term's grand total; it is how many mappings *this search, with your current options*, will actually show. A suggestion is therefore a promise that the results table will not be empty, and it narrows as you narrow the search: choosing **From ontologies**, or leaving the weak-predicate boxes unticked (§ 3.6), removes terms from the list rather than offering you a dead end.

3.4 Loading a list of terms from a file

In multiple-term mode you can drop a `.txt`, `.csv` or `.tsv` file (up to 1 MB) onto the dashed box, or click to pick one. The file is read **in your browser** — nothing is uploaded anywhere — and its terms are *appended to the text area*.

That last detail is deliberate. The text area is always the single, visible, editable list of what will be searched. A file never becomes a hidden second source of truth that you cannot inspect. Terms already in the box are not added twice.

3.5 Between which ontologies

Two optional selectors, **From ontologies** and **To ontologies**, restrict the source and target of the mappings. Each option is annotated with how many mappings it takes part in. The ⇄ button between them swaps the two sides.

They compose with the term box in three useful ways:

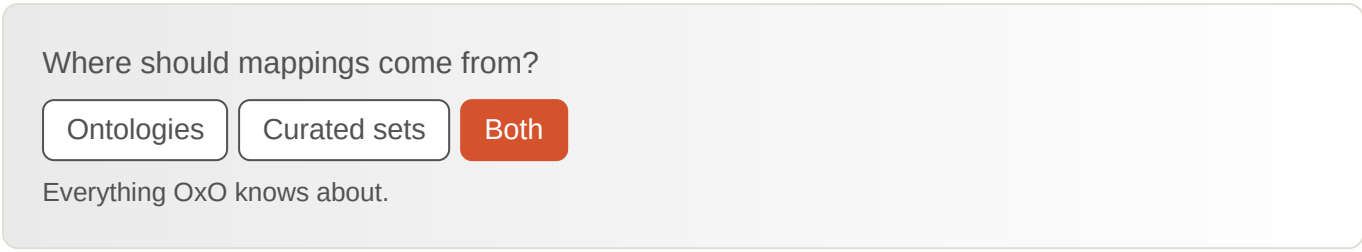
- **Terms only** — every mapping involving those terms.
- **Terms + a target** — a bounded batch lookup: "map these 400 terms into MONDO".
- **A source, and an empty term box** — the whole ontology: "everything DOID maps to".

Pick exactly one source and the target list narrows to the ontologies actually reachable from it, with their counts, so you cannot pick a target that will return nothing.

3.6 More options

Everything else lives behind ▸ **More options**, in three groups: *where mappings come from*, *what else to show*, and *how labels match*. When the disclosure is closed, any choice you have changed from its default is summarised in brackets on the summary line — so a forgotten restriction cannot silently narrow your results.

Where should mappings come from?



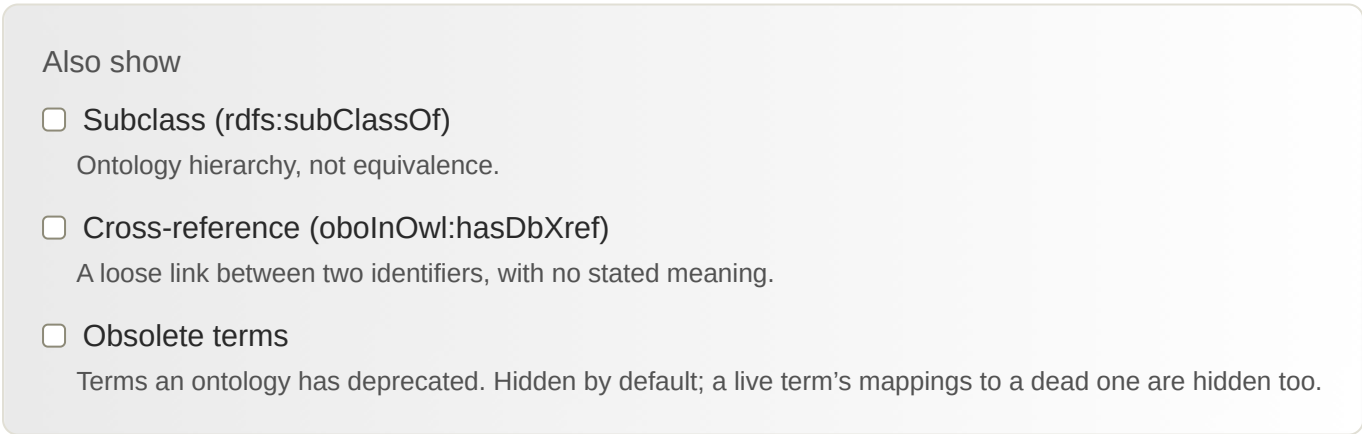
The corpus control. The hint underneath changes with your choice.

This is the **Ontologies** / **Curated sets** distinction from § 1. **Both** is the default. Choose deliberately: a cross-reference an ontology asserts about its own terms and a mapping a curator drew between two projects are different kinds of evidence, and which you want depends on what you are doing.

> **THIS CONTROL DOES NOT HIDE INFERRED MAPPINGS**
> An inferred mapping chains premises from several sets at once, so it belongs to no single corpus and always survives this filter. Hiding inferences is a separate control, on the results table (§ 4.4).

Beneath it, **Choose specific mapping sets…** opens a picker of every set OxO holds, as **two tables**: *Curated sets*, with columns Title, Mapping Set Id, Description and Provider; and *Ontologies*, with columns Ontology, Prefix and Obsolete. Both have per-column filter boxes and sortable headers. Click anywhere on a row to toggle it. With nothing selected, all sets are searched.

Also show



All three are off by default. This is the most consequential group in the form.

The first two are the **weak predicates**. `rdfs:subClassOf` says one term is *narrower* than another, not equal to it; `oboInOwl:hasDbXref` says two identifiers are related somehow, without saying how. Neither is a mapping between equals, and both are extremely numerous, so by default OxO leaves them out and answers with mappings that make stronger claims, such as equivalence, exact match and same as.

> **THIS IS WHY A SEARCH CAN COME BACK EMPTY**
>
> On an ontology-derived corpus the overwhelming majority of cross-references are `hasDbXref`. A term whose every mapping is a cross-reference therefore returns **nothing at all** by default — and, because the suggestion list honours the same setting, it will not even be offered as a suggestion. If a term you know exists finds nothing, tick **Cross-reference** first.

Obsolete terms are terms an ontology has deprecated. Hiding them is the default because a mapping to a dead term is usually not the answer you want. The switch governs the results table, the suggestion list and the mapping-set picker together, and it is inherited by inferences: an inferred mapping that passes through an obsolete term is hidden with it.

Label matching

Controls how free-text terms match, and nothing else — CURIEs and IRIs are always exact lookups.

- **Exact match (ignore case)** — the default. `cataract` finds *Cataract*.
- **Partial match** — substring-ish. Broad; expect volume.
- **Exact match (case-sensitive)** — for when case is meaningful.

3.7 Search and Clear

Clear resets the whole form — terms, mapping sets, ontologies, corpus, weak predicates, obsolete terms and label matching — back to defaults. On the results page it also drops every column filter and sort from the results table, so one button really does mean “start again”. The one thing it leaves alone is the Single/Multiple term choice.

Searching with neither terms nor ontologies does nothing, silently. There is nothing to search for.

4 Reading the results

4.1 The toolbar

Above the table sit a count — **“Mappings found: *n*”**, with the ontologies appended when you restricted them (*“· from MONDO → EFO”*) — and a **download** icon (§ 4.7). At the far right is a button to **show and hide columns**, which also lets you drag them into a different order.

> **THE COUNT COUNTS ROWS, NOT MAPPINGS**
>
> Because identical triples are collapsed into one row (§ 4.3), the number is how many *distinct* subject–predicate–object statements matched, not how many mapping records exist behind them. On a mapping set page (§ 8), where nothing is collapsed, it is the raw count.

4.2 The columns

Column	Shows
Subject, **Object**	Label (bold), then identifier, then IRI. The identifier and IRI each have a copy button, and there is an OLS link that opens the term in a new tab. A term with no label leads with its identifier instead.
Predicate	Same three lines, no OLS link. If the mapping carries a predicate modifier, a red NOT badge appears in front of it — that mapping asserts the *negation*. Do not skim past it.
Mapping justification	Why the mapping is claimed to hold, as the short SEMAPV name — `ManualMappingCuration`, `LexicalMatching`, `MappingChaining`. Hover for the full label.
Mapping confidence	The confidence the publisher declared, as published. An em dash means they declared none.
Distance	How many ontologies the mapping spans (§ 9). `1` for anything asserted.
Type	Asserted or SSSOM inference.
Mapping provider	Who supplied it. This field is *hidden by default* because most of the datasets do not populate it currently. You can turn it on with the columns button.
Mapping set	The set’s title, linking to its own page (§ 8), with the identifier and a copy button beneath.

The eye icon at the end of a row opens the mapping detail page (§ 6).

4.3 One row can be several mappings

Several records may describe the same subject, predicate, predicate modifier and object. OxO collapses them into one **expandable** row, using the highest-scoring matching record as its representative. A negated statement remains separate from its positive counterpart. The "*n* matching records · View all support" badge counts records matching your current search. The chevron expands up to 21 matching records, each with its own type, justification, confidence, distance, set, and eye icon. If there are more, use **View all supporting mappings** to reach the complete, paginated list on the mapping detail page (§ 6).

View all support includes records across the whole dataset, even those outside your search filters. Its count can therefore exceed the matching-record count. Several records can share one justification method or come from one mapping set; these are different counts.

Where the collapsed members disagree on a field, the cell reads *Multiple* (or *Multiple sets*) rather than misleadingly showing one of them.

4.4 Narrowing what you see

The **funnel icon** in a column header opens a filter. Subject, Predicate and Object each offer three boxes — *ID*, *label* and *IRI* separately; Mapping justification, Distance and Mapping provider offer one. Filters are combined with AND across every column, apply about 400 ms after you stop typing, and put a coloured dot on the funnel when active. Each popover has its own **Clear**.

Typing into a filter matches on *contains*. Several of the filters also offer a list of the values actually present in your current results, with counts; **picking** from that list matches that value *exactly*. So a suggested value can never empty the table, and typing over a picked one relaxes it back to a substring match.

The **Type** filter is a three-way choice of *All* (the default), *Asserted* only, or *SSSOM inference* only. **This is the control that hides inferred mappings** — not the corpus selector.

The **Distance** filter is a single *Max distance* dropdown: picking 2 means "at most 2". Its choices adapt to what is actually in your results — if nothing reaches past 3, it offers 1 to 3 — and are capped at 5, with an *All* entry appearing only when the cap actually bites.

4.5 Re-ordering

The **sort icon** on Subject, Predicate and Object lets you sort that column by ID, label or IRI, ascending or descending. Sorting is single-key *within* a column — choosing IRI replaces label — but multi-column *across* them: sort by Subject, then Object, and each icon shows a small number giving its place in the sort order. **Clear** removes just that column from the sort; clearing them all returns to the default order.

Remember § 5: the moment you sort by anything, relevance ranking stops applying.

4.6 Paging

Page size is 5, 10 (the default), 20, 30, or 50 rows.

4.7 Export

The **download icon** exports the *entire* result set — not just the page you are looking at — as SSSOM-compliant TSV, named `oxo2-mappings.tsv`. Every filter you have applied is honoured.

Two deliberate differences from what is on screen:

- **Rows are not collapsed.** The file has one line per mapping, so a triple asserted by four sources appears four times. This is the SSSOM-faithful form.
- **Your sort is not applied.** The export is a data set, not a view of one.

4.8 The address bar is the state

Your terms, ontologies, corpus, label matching, weak predicates, obsolete setting, page, page size, sort and every column filter live in the URL. Three consequences worth knowing:

- A results URL **describes the view completely**, so it survives a reload and travels to a colleague. It carries the query, not the data — on a different instance, or after a reindex, the same URL can return different rows.
- Opening a mapping's details and pressing **Back to results** returns you to the same page, sort and filters — not to page 1.
- Defaults are omitted, so an untouched search has a clean URL.

Which columns you have shown or hidden is the exception: it is not in the URL, so it is not shared with the link and does not survive a reload.

5 How results are ordered

By default, **mappings with more direct asserted source sets come first**. Three source sets rank above two, two above one, and one above zero. When the count is equal, OxO uses its relevance score, which combines how well your query matched with provenance, predicate strength, manual curation and declared confidence.

FIRST · DIRECT ASSERTED SOURCE SETS

More distinct source sets first: 3 sources › 2 sources › 1 source › 0 sources

THEN · RELEVANCE SCORE

Query match × provenance × predicate strength × manual curation × confidence

Source count takes priority. The relevance score orders mappings with the same count.

Direct support counts distinct mapping sets asserting the **same subject, predicate, modifier and object** across the whole dataset. Search filters do not change that count. Repeated records from one set and additional justification methods do not increase it. Inference premises that make other statements do not count as direct sources for the conclusion.

Example result	Direct sources	Default position
Asserted by three curated mapping sets	3	First
Asserted by two curated mapping sets	2	Second
Asserted by one ontology mapping set	1	Third
Inferred, with no direct asserted source	0	Fourth

Among results matching the search, source count wins even over a higher-provenance or closer text match.

Different source sets can reuse the same evidence. Their count is a corroboration signal, not a confidence estimate or a count of independent confirmations. Missing source counts are treated as zero. **Recency is not a ranking factor.**

When the source count is equal

The query relevance score is multiplied by four factors:

Factor	Multiplier
Provenance	Ontology assertion ×10 000; curated or unknown-category assertion ×1 000; inference ×100, divided by 5 per extra ontology spanned.
Predicate strength	OWL identity ×2; exactMatch ×1.7; crossSpeciesExactMatch ×1.6; closeMatch ×1.4; broad/narrow ×1.2; other ×1.
Manual curation	ManualMappingCuration ×1.3; other methods ×1.
Confidence	×(1 + 0.3 × confidence); no declared confidence ×1.

At equal source count and query relevance, provenance dominates: the maximum curated boost (1 000 × 2 × 1.3 × 1.3 = 3 380) remains below the minimum ontology-asserted boost (10 000). The other factors blend together. A closer text match can outweigh provenance when source counts are equal; it cannot outweigh a higher source count.

> **RANKING IS NOT VISIBILITY**
> Ranking orders the mappings that match your search; it does not make hidden mappings eligible. Weak predicates and obsolete terms remain hidden unless you enable them (§ 3.6), regardless of how many sources support them.

> **SORTING TURNS RANKING OFF**
> A column sort replaces both source-count ordering and the relevance score. Clear every column sort to return to the default order (§ 4.5).

6 The mapping detail page

Reached from the eye icon on any result row — or from its own URL: `/mapping/<id>` is a real permalink, so you can bookmark it, share it, and refresh it. The page lays out everything SSSOM records about the mapping, in sections that hide themselves when empty, so what you see is what the source actually published:

- **Mapping** — the triple, rendered large, label above identifier.
- **Supporting mappings** — all records for the same statement, with counts and links to their details (below).
- **Inference Details** — for inferred mappings only: an *Asserted Mappings* table of the premises it rests on, and the explanation graph (§ 7).
- **Justification** — mapping justification and confidence.
- **Provenance** — author, creator, reviewer, mapping date, publication date, provider.
- **Mapping Set** — the set's identifier, linking to its page (§ 8), plus title, description, source and version.
- **Additional Information** — licence, comment, mapping tool and version.

- **Subject Details** / **Object Details** — category, type, source, source version.

Metadata sections have a **Hide** / **Show** toggle on their heading. The button at the top right reads **Back to results** when you arrived from a search — and returns you to it with your filters and page intact — or **New search** when you opened the page cold.

Supporting mappings

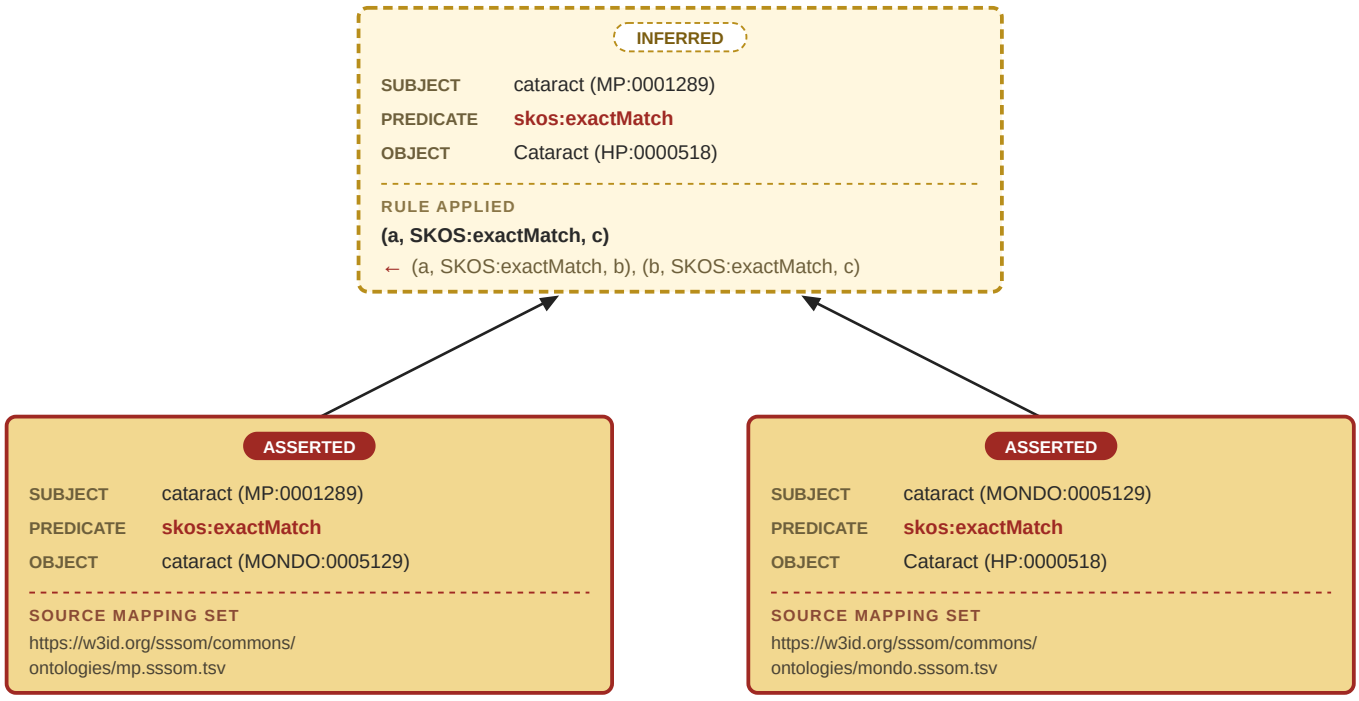
This section lists the current record alongside every other record with the same subject, predicate, modifier and object. It shows the total number of records, the number of distinct nonblank justification methods, and the number of distinct nonblank source-set identifiers among the asserted records. Counts cover the full group, not just the current page. Use the page controls to see more records and each record's link to inspect its details.

The list includes records hidden by your search filters. For an inferred mapping, the section appears only when there is another supporting record besides the current one. An asserted singleton still shows its support summary. If loading fails, use **Retry**.

These supporting records make the *same statement*. The asserted premises used to derive an inference usually make different statements; those remain in **Inference Details** and its explanation graph (§ 7). An inference can therefore have premises but zero direct asserted sources.

7 Explaining an inferred mapping

When OxO derived a mapping rather than reading it somewhere, it can show you exactly how. The detail page of an inferred mapping ends with a graph: **arrows point from premises to the conclusion**, so the mappings somebody actually asserted sit at the bottom and the derived one sits at the top.



Illustrative shape, not a real result. Real chains can be deeper and wider.

Reading the graph:

- **Solid box, dark red border,** ASSERTED **badge** — a mapping somebody published. Its footer, headed *Source mapping set*, gives the full identifier of the set it came from, which is how you see that a single inference crosses several sources.
- **Dashed box, amber border,** INFERRED **badge** — a mapping OxO derived. Its footer, headed *Rule applied*, names the SSSOM chaining rule that produced it, written as `conclusion ← premises`, with `a`, `b`, `c` standing for the terms being chained.
- Every arrow points **premise → conclusion**. Follow them upwards to see how the claim was built; follow them down to reach evidence a human wrote.

Drag to pan, scroll to zoom, and use the buttons in the corner to zoom or fit the whole graph. The nodes are diagrams, not links.

8 Mapping set pages

Every mapping set name in the interface is a link to that set's own page. It opens with the set's title and type badge, its description, its full identifier, and its provider and creators where they are published — then lists all of the set's mappings in the usual results table.

Two differences from a search results table, both because every row belongs to the same set:

- The **Type** and **Mapping set** columns are dropped — they would repeat the same value on every row.
- Rows are **not collapsed** by identical triple, so the count is a straight count of mappings and there is nothing to expand.

OxO's own inferences are published as mapping sets with resolvable identifiers, so `/inferences` is one of these pages rather than a dead identifier — as is `/inferences/<source>` for the inferences drawn from a particular source set. They show every inference they hold regardless of the usual type filter: you navigated to a set of inferences, so hiding inferences would be perverse.

9 Glossary

Asserted mapping
A mapping somebody published, in an ontology or in a curated SSSOM file. Badged Asserted.

Inferred mapping (SSSOM inference)
A mapping OxO derived by chaining asserted ones. Badged SSSOM inference. Always explainable (§ 7).

Mapping set
A published collection of mappings with its own identifier, title, licence and version.

Mapping set category
Whether a set is an *ontology*'s own cross-references or a *curated* SSSOM file. Drives the corpus control (§ 3.6) and the top ranking tier (§ 5). It is an OxO judgement about the source, not something SSSOM records.

CURIE

A compact identifier: a prefix, a colon, a local name — `MP:0001289`. Expands to an IRI.

Distance

How many ontologies a mapping spans, counted over its whole explanation: one fewer than the number of distinct CURIE prefixes involved, and never less than 1. An asserted mapping, or an inference that stays within two ontologies, is 1; three ontologies is 2. Each step up divides the ranking by five.

Mapping justification

The SSSOM field saying *why* a mapping holds — hand-curated, lexically matched, logically reasoned, and so on. Shown by its short SEMAPV name, with the full label on hover.

Obsolete term

A term its ontology has deprecated. Hidden by default, along with any mapping — asserted or inferred — that touches one (§ 3.6).

Predicate modifier

A negation. A mapping carrying one asserts the triple does *not* hold; the table shows a red NOT badge.

SSSOM

The Simple Standard for Sharing Ontological Mappings — the vocabulary and TSV format OxO reads, stores and exports.

Weak predicate

`rdfs:subClassOf` and `oboInOwl:hasDbXref`: links that do not claim two terms are equivalent. Excluded from results by default; each has its own checkbox (§ 3.6).

10 Things that will surprise you

1. **A term you know exists returns nothing.** Weak predicates are hidden by default, and on ontology-derived data most cross-references are `oboInOwl:hasDbXref`. Tick **Cross-reference** (and, if needed, **Subclass**) under *More options*. The suggestion list hides the same terms, so their absence there is a symptom of the same setting, not evidence the term is missing. See § 3.6.

2. **Obsolete terms are hidden, including from live terms' results.** A mapping from a current term to a deprecated one does not appear until you tick **Obsolete terms**. Inferences that route through an obsolete term vanish with it.

3. **Your term is only matched as the subject.** Mappings where it is the object are not returned. Search the other term, or swap the From/To ontologies. See § 3.2.

4. **Sorting a column silently disables relevance ranking.** There is no "sort within relevance". See § 5.

5. **A red NOT badge inverts the row's meaning.** It is small and it is easy to skim past, and the mapping says the opposite of what the row appears to say.

6. **The result count is not the number of mappings.** On a search it counts collapsed triples; on a mapping set page it counts mappings. The download always contains the latter.

7. **Shown/hidden columns are not part of the shareable URL.** Everything else about your view is. Reveal *Mapping provider*, send someone the link, and they will not see it.

OxO2 is developed by the Samples, Phenotypes and Ontologies Team (SPOT) at EMBL-EBI. Report a problem, or ask for a feature, at github.com/EBISPOT/oxo2/issues.

This document is generated from `docs/user/ui/oxo2-user-interface.html` in the OxO2 repository. The behaviour it describes is defined by the architectural decision records in `docs/adr/` — chiefly ADR-0011 (inference type), ADR-0013 (same-triple grouping), ADR-0024 (cross-ontology mapping), ADR-0026 (label matching), ADR-0027 (mapping set category and provenance-led ranking), ADR-0030 (subject-side search), ADR-0031 (distance as ontology span), ADR-0034 and ADR-0035 (suggestions and weak predicates), ADR-0036 (search options grouped by intent), ADR-0038 (the mapping set picker), ADR-0040 (removal of Advanced search) and ADR-0041 (obsolete terms), and ADR-0054 (supporting mappings and corroboration ranking).